

Chapter 1 – Introduction to Computer Networking (Detailed Exam Notes)

1. Introduction

Network:

A collection of interconnected elements or nodes that communicate and interact with each other to achieve a specific purpose or to facilitate the exchange of information, resources, or services. Networks can take various forms and serve different functions.

Computer Network:

A system of interconnected computers and devices that enable data and information sharing, such as the internet or a local area network (LAN)

2. What is the Internet?

Two main views:

2.1 Nuts-and-Bolts View

The Internet consists of billions of connected computing devices, known as hosts or end systems, running network applications at the "edge". These devices are connected through packet switches (routers and switches) and various communication links (fiber, copper, radio, satellite). The Internet is a "network of networks" of interconnected ISPs, relying on protocols like HTTP, TCP, IP, and WiFi for message control.

- Internet = **network of networks**.
- **Billions** of connected computing devices:
 - *Hosts / End Systems* (e.g., computers, smartphones).
 - Run **network applications** (e.g., web browsers, email clients).
- Devices connected via:
 - **Communication links**: fiber optic, copper wire, radio, satellite.
 - Measured in **bandwidth** (bits per second).
 - **Packet switches**: forward packets through the network.
 - Examples: **routers, switches**.

2.2 Service View

The Internet provides infrastructure for various applications, including web, streaming video, e-commerce, and social media. It offers a programming interface with "hooks" for applications to connect and use Internet transport services, similar to postal service options.

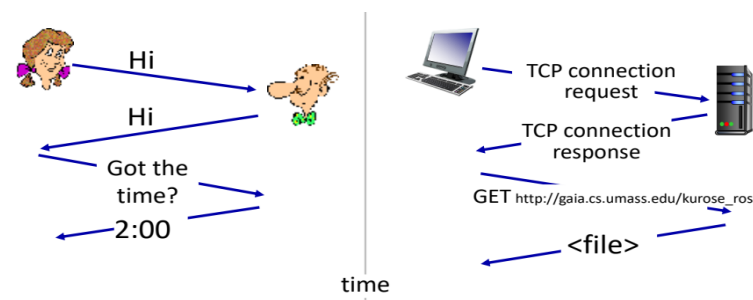
- Internet provides infrastructure for **applications** like:
 - Web, video streaming, email, online games, social media.
- Applications use **Internet protocols** to send/receive data.
- Examples of protocols: HTTP, TCP, IP, WiFi, Ethernet, 4G.

3. What is a Protocol?

A **protocol** is a set of rules that computers follow to communicate with each other, just like people follow a language to talk. It defines how data is formatted, sent, received, and handled for errors. Examples include **HTTP** for web browsing, **TCP/IP** for reliable data transfer, and **DNS** for finding website addresses.

A **protocol** defines:

- The **format** of messages.
- The **order** they are sent and received.
- The **actions** taken when a message is received.
- **Human Analogy:**
 - Saying "Hello" and waiting for "Hello" back before talking — that's a protocol.
- **Networking Examples:**
 - HTTP: Defines how browsers and servers exchange web pages.
 - TCP: Ensures reliable delivery of data.
 - WiFi: Defines how devices communicate over a wireless network.



4. Network Structure

The **network edge** is made up of devices like clients (users' computers or phones) and servers (often stored in data centers). These devices are connected through **access networks** and **physical media**, such as wired cables or wireless links. The **network core** is the center part of the internet, made up of many routers connected together, forming a big "network of networks."

4.1 Network Edge

- **Hosts:**
 - Clients (request services) and Servers (provide services).
 - Servers often located in **data centers**.
- **Access Networks & Physical Media:**
 - **Wired:** DSL, Cable, Ethernet.
 - **Wireless:** WiFi, 4G/5G.

- **Connection to Edge Router:**
 - Through ISP access networks.

4.2 Network Core

- Mesh of interconnected **routers**.
- **Packet Switching:**
 - Breaks messages into **packets**.
 - Packets travel router-to-router.
- **Circuit Switching:**
 - Reserves a dedicated path for the entire communication.

5. Access Networks and Physical Media

5.1 Connecting End Systems to the Internet

An access network is a telecommunications network that connects individual users or devices to their immediate service provider. End systems (like laptops, smartphones, desktops) connect to the Internet through **access networks**.

The main types are:

1. **Residential Access Networks** – Used in homes (e.g., DSL, Cable Modem, Fiber).
2. **Institutional Networks** – Used in schools, universities, companies.
3. **Mobile Access Networks** – Wireless networks provided by cellular operators.

When evaluating an access network, two factors matter:

- **Transmission rate:** The speed of data transfer, measured in **bits per second (bps)**.
- **Shared or Dedicated Access:** Whether multiple users share the same connection (shared) or each has their own bandwidth (dedicated).

5.2 Cable-Based Access

- Uses **Hybrid Fiber Coax (HFC)**:
 - Combines optical fiber and coaxial cable.
 - Data and TV signals are sent at different frequencies.
- **Downstream:** Up to **1.2 Gbps**.
- **Upstream:** 30–100 Mbps.
- Multiple homes share the same cable segment, so performance may drop at peak hours.

- Uses **Frequency Division Multiplexing (FDM)** — each service gets a separate frequency band.

5.3 Digital Subscriber Line (DSL)

- Uses **existing telephone lines** to connect to the Internet.
- Data goes through a **DSL modem** to the **DSL Access Multiplexer (DSLAM)** at the telephone company's central office.
- Voice and data are transmitted at different frequencies, so you can use the phone and Internet at the same time.
- **Speeds:**
 - Downstream: 24–52 Mbps.
 - Upstream: 3.5–16 Mbps.
- Unlike cable, each user has a dedicated line to the central office — no sharing with neighbors.

5.4 Home Networks

A typical home network has:

- A **modem** (Cable or DSL) connected to the ISP.
- A **router** (often combined with firewall and NAT functions).
- Wired Ethernet connections (1 Gbps).
- **WiFi access point** for wireless devices.

5.5 Wireless Access Networks

Two main types:

1. **Wireless Local Area Networks (WLANs)** –
 - Typically inside a building.
 - Uses IEEE 802.11 (WiFi) standards (11 Mbps, 54 Mbps, 450 Mbps).
2. **Wide-Area Cellular Networks** –
 - Provided by mobile network operators.
 - Covers long distances (tens of km).
 - 4G (~10 Mbps) and 5G (much higher speeds).

5.6 Enterprise Networks

- Found in organizations (universities, companies).
- Mix of wired and wireless connections.
- Wired Ethernet: 100 Mbps to 10 Gbps.
- Wireless access points for mobility inside the organization.
- Connected to the Internet through institutional routers.

6. Host Operations and Link Concepts

6.1 Hosts

- **Sending function:** Takes an application message, breaks it into **packets** (L bits each), and sends them into the network at a given **transmission rate (R)**.
- **Transmission delay:** Time to push all L bits into the link.
 - Formula: **Delay = L / R**.

6.2 Physical Media

- **Guided media:** **Signals travel through a solid medium.**
 - **Twisted pair:** Two copper wires twisted to reduce interference. Cat 5 (1 Gbps) and Cat 6 (10 Gbps).
 - **Coaxial cable:** A coaxial cable is a type of guided transmission medium that carries signals through a central copper wire surrounded by insulation, a metallic shield, and an outer cover. It is commonly used for cable TV, internet connections, and CCTV cameras(100 mbps)
 - **Fiber optic:** Fiber optic cable uses glass fibers to carry light pulses, enabling high-speed point-to-point transmission (10's-100's Gbps) with low error rates and immunity to electromagnetic noise
- **Unguided media:** **Unguided media** means communication that happens **without physical cables or wires**. Instead, signals travel through the **air or space** using electromagnetic waves.
 - **Terrestrial microwave:** Up to 45 Mbps per channel.
 - **WiFi:** Hundreds of Mbps.
 - **Cellular:** 4G (~10 Mbps), 5G (much higher).
 - **Satellite:** Covers large areas but has high delay (up to 45 Mbps, 270 msec delay).

7. The Network Core

The **network core** is the central part of the Internet that connects different networks together. It consists mainly of **interconnected routers** and **high-speed links**. Its main role is to **forward packets** from the source to the destination through a sequence of routers.

7.1 Packet Switching

- In **packet switching**, the message is divided into **small packets** before being sent.
- Each packet travels independently through the network.
- At each router, the **entire packet** must arrive before it can be forwarded — this is called the **store-and-forward** principle.
- **Transmission delay** formula:

$$Delay = L/R$$

where **L** = packet length in bits, **R** = transmission rate in bits/sec.

Example:

If **L** = 10 Kbits and **R** = 100 Mbps,

Delay = 0.1 milliseconds for one hop.

Store-and-Forward Transmission in Packet Switching

In packet-switched networks, data is divided into **packets**. Each packet must be **fully received** by a router **before** the router can begin transmitting it to the next link in the path. This process is called **store-and-forward**.

- **Store**: The router temporarily stores the entire packet in its memory buffer.
- **Forward**: After the packet is completely received, the router forwards it to the next hop toward the destination.

End-to-End Delay Calculation (Ignoring Propagation Delay): End-to-End Delay Calculation means finding out how long it takes for data to travel from the sender (source) to the receiver (destination) across a network.

If there are two links between the sender and the receiver, each with the same transmission rate **R** (in bits per second), and the packet length is **L** (in bits), then:

- Time to transmit the packet over the **first link** = **L / R** seconds.
- Time to transmit the packet over the **second link** = **L / R** seconds.

Therefore,

$$\text{Total end-to-end delay} = L/R + L/R = 2L/R$$

Example for exam:

If **L** = 1,000 bits and **R** = 500 bits/sec,

Delay = $2 \times (1000/500) = 4$ seconds.

7.2 Queueing Delay and Packet Loss

Queueing Delay:

- When packets arrive at a router or switch faster than they can be sent out, they wait in a **queue (buffer)**.
- The time a packet spends waiting in this queue is called **queueing delay**.

- More traffic → longer queues → higher queueing delay.

Packet Loss:

- If the queue is **full** and more packets arrive, the router **drops the extra packets**.
- Dropped packets are called **lost packets**, which may need **retransmission** depending on the protocol (like TCP).
- Packet loss usually happens because of **network congestion**.

7.3 Forwarding vs Routing

- **Forwarding:** Forwarding involves moving arriving packets from a router's input link to the appropriate output link based on the destination address in the packet's header.
- **Routing:** Routing determines the source-destination paths taken by packets using routing algorithms.

7.4 Alternative to Packet Switching: Circuit Switching

Circuit switching establishes a dedicated communication path between the source and destination for the duration of a “call.” It allocates and reserves end-to-end resources, ensuring consistent and guaranteed performance.

However, the reserved circuit remains idle when not in use, as resources are not shared. Circuit switching is commonly employed in traditional telephone networks due to its reliability and predictable performance.

- Two main types:
 1. **Frequency Division Multiplexing (FDM)** – Each user gets a separate frequency band.
 2. **Time Division Multiplexing (TDM)** – Each user gets a periodic time slot.

Advantages:

- Guaranteed bandwidth.
- Predictable performance.

Disadvantages:

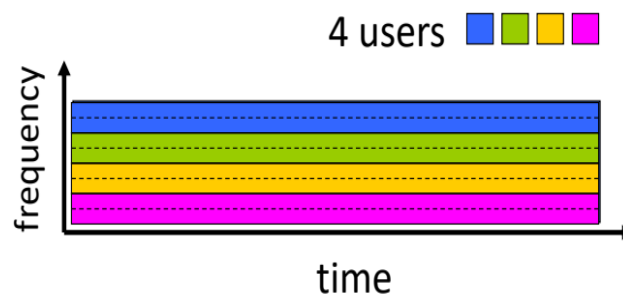
- Resources are idle when not in use.
- Less efficient for bursty data traffic.

Circuit Switching: FDM and TDM

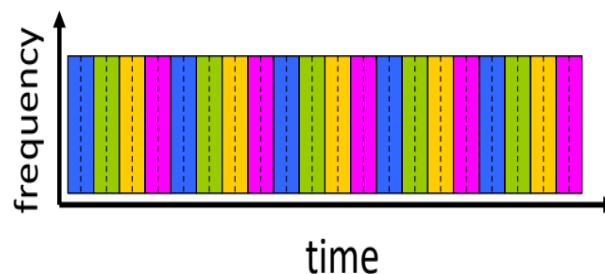
Circuit switching can use **Frequency Division Multiplexing (FDM)** or **Time Division Multiplexing (TDM)** to allow multiple users to share a communication channel.

In **FDM**, the available frequency spectrum is divided into narrow **frequency bands**, and each call gets a dedicated band. Each user can transmit continuously at the **maximum rate** of its band. FDM is

common in analog systems, such as traditional telephone networks and radio communication. However, if a call is idle, its frequency band remains unused, which can reduce efficiency.



TDM, on the other hand, divides the channel in **time slots**. Each call is allocated periodic slots, during which it can use the full bandwidth of the channel. This allows higher data rates than FDM but requires careful synchronization to avoid overlap. TDM is widely used in digital telephony and networks.



Both FDM and TDM have their advantages: FDM is simple and continuous, while TDM maximizes bandwidth usage by sharing it in time.

7.5 Packet Switching vs Circuit Switching

Circuit Switching establishes a **dedicated communication path** between the sender and receiver for the duration of a call. It reserves **end-to-end resources**, ensuring **consistent and guaranteed performance**. Examples include traditional telephone networks. However, the dedicated path remains **idle when not in use**, leading to **inefficient utilization** of network resources. Circuit switching is ideal for applications requiring **steady, continuous transmission**, such as voice calls.

Packet Switching, in contrast, divides data into small **packets**, which are sent independently over the network and may follow **different paths** to reach the destination. Resources are **shared among multiple users**, making the network **more efficient** and scalable. Packet switching is the foundation of modern data networks, including the Internet, as it supports **bursty traffic** and **dynamic routing**. Unlike circuit switching, packet switching does not guarantee fixed delays or bandwidth, but it allows **better overall utilization** of network resources and flexibility for various types of data.

8. Internet Structure – A Network of Networks

8.1. Hosts and Access ISPs

- A **host** is any device on the Internet — like your phone, laptop, or a web server.
- To connect to the Internet, a host needs an **access ISP** (Internet Service Provider) — this could be your local broadband company or mobile carrier.

8.2. Why ISPs Must Interconnect

ISPs (Internet Service Providers) must **connect with each other** so that their users can communicate with anyone on the internet, not just within their own network. Without interconnection, a user on one ISP couldn't send emails, browse websites, or access services hosted on another ISP.

- **Reason 1:** To **exchange traffic** efficiently between networks.
- **Reason 2:** To ensure **global internet connectivity** — making the internet one big network instead of many isolated ones.
- With **millions** of access ISPs worldwide, directly connecting each ISP to every other ISP would require an impossible number of links — about **$O(N^2)$** connections — meaning the number of links grows very fast as the number of ISPs increases.

8.3. One Simple (But Impractical) Idea

Even if a single global ISP would make things simple, competition and multiple providers are inevitable. As a result, ISPs must interconnect with each other so that all users, no matter which ISP they use, can communicate.

8.4. How ISPs Actually Interconnect

- ISPs connect in different ways depending on business deals and national policies.
- Two important ways they connect:
 1. **Internet Exchange Points (IXPs)** — large network hubs where many ISPs connect their networks in one place.
 2. **Peering Links** — direct agreements between two ISPs to exchange traffic without paying a third party.

8.5. Regional and Content Provider Networks

- **Regional Networks:** Smaller ISPs often connect to **regional ISPs** that provide a bigger coverage area (like an ISP for a whole country or large geographic area).
- **Content Provider Networks:** Big companies like **Google, Microsoft, and Akamai** run their own global networks so they can deliver their services and data faster — sometimes bypassing traditional ISPs.

At the core of the Internet:

- **Tier 1 ISPs:** These are the biggest internet providers with networks all over the world. They connect with each other to make sure data can travel anywhere globally.

9. Packet Loss and Delay

9.1 Introduction

- In computer networks, **packet loss** and **delay** occur mainly due to congestion and limited resources in routers and links.
- When packets travel between a sender (A) and a receiver (B), they pass through routers where they may wait in buffers before being transmitted.

9.2 Packet Queuing in Router Buffers

1. Packets Queue and Wait

- Packets arriving at a router are placed in a **buffer** (queue) if the output link is busy.

2. Cause of Packet Loss

- If the **arrival rate** of packets to a link temporarily exceeds the **output link capacity**, the buffer may fill up.
- When **no free buffers** are available, newly arriving packets are **dropped** → **packet loss**.

9.3 Packet Delay and Loss in Computer Networks

1. Packet Delay

When a packet travels from source to destination, it experiences several types of delays:

(a) Processing Delay

Processing delay is the time a router or host takes to look at a packet's header, check for errors, and decide where to send it next

(b) Queueing Delay

Queueing delay is the time a packet spends waiting in line (buffer) at a router or switch before it can be sent.

(c) Transmission Delay

Transmission delay is the time it takes to push all the packet's bits onto the communication link.

Formula:

Transmission Delay = Packet Size (bits) / Link Bandwidth (bps)

$$d_{\text{trans}} = L/R$$

where:

- L = packet length (bits)
- R = link transmission rate (bps)

Example: Sending a 1,000-byte packet over a 1 Mbps link = $(8000/10^6) = 8$ ms.

(d) Propagation Delay

Propagation delay is the time it takes for a signal to travel from the sender to the receiver through the physical medium (like a cable or fiber).

Formula:

Propagation Delay = Distance / Propagation Speed

$$d_{\text{prop}} = d/s$$

where:

- d = physical link length (meters)
- s = propagation speed ($\sim 2 \times 10^8$ m/sec in fiber)

Example: 1,000 km over fiber ($\approx 2 \times 10^8$ m/s) $\rightarrow 5$ ms.

Total Nodal Delay = Processing + Queueing + Transmission + Propagation

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

2. Packet Loss

Packet loss happens when one or more packets of data **fail to reach their destination** in a network.

Causes of Packet Loss:

Packet loss can happen due to several reasons:

1. **Network Congestion:** Too many packets arrive at a router or switch, filling up its buffer, so extra packets are dropped.
2. **Buffer Overflow:** When a device's memory (buffer) for holding packets is full, any new incoming packets are discarded.
3. **Transmission Errors:** Noise or interference on the physical medium (cables, wireless) can corrupt packets, causing them to be discarded.
4. **Faulty Hardware:** Routers, switches, or network cards can fail, leading to dropped packets.
5. **Software Issues:** Bugs in network devices or misconfigured settings can result in packet loss.
6. **Routing Problems:** Incorrect or unstable routing paths can cause packets to be lost in transit.

3. Relation Between Packet Loss and Delay

High traffic load $\rightarrow$ high queueing delay $\rightarrow$ buffers fill $\rightarrow$ packet loss occurs.

If arrival rate $>$ link capacity for a long time $\rightarrow$ congestion collapse.

10. Throughput

(a) Definition

- **Throughput:** Throughput is the actual rate at which data is successfully transmitted from sender to receiver over a network.

(b) Types of Throughput

1. **Instantaneous Throughput:** Rate at a specific instant of time.
2. **Average Throughput:** Rate measured over a longer duration.

(c) End-to-End Throughput

Simple Pipe Model

- Think of data flow as **fluid through a pipe**.
 - **Sender rate (R_s):** Data generation speed at sender (bits/sec).
 - **Link capacity (R_c):** Maximum transmission speed supported by the network (bits/sec).
1. **Case 1:** If $R_s < R_c \rightarrow$
 - Sender is slower than the link.
 - **Throughput = R_s .**
 2. **Case 2:** If $R_s > R_c \rightarrow$
 - Link is the limiting factor.
 - **Throughput = R_c .**

Conclusion: End-to-end throughput = $\min(R_s, R_c)$.

(e) Bottleneck Link

- **Definition:** A bottleneck link is the slowest part of a network path that limits the overall data transfer rate
- Example:
 - Multiple links between sender and receiver with different capacities.
 - Effective throughput = **capacity of the bottleneck link**.

(f) Network Scenario (Multiple Connections)

- Suppose **10 connections** share a single bottleneck link of capacity **R bits/sec**.
- **Per-connection throughput** $\approx R/10$ (fair share).

12. Network Security

1. Introduction to Network Security

- **Definition:** Network security deals with protecting computer networks against unauthorized access, misuse, malfunction, modification, destruction, or improper disclosure.
- **Main fields of study:**
 1. **Attacks** → How attackers (“bad guys”) exploit networks.
 2. **Defense** → How to secure and defend against attacks.
 3. **Design** → Building secure architectures resistant to attacks.

2. Types of Threats & Attacks

(a) Malware (Malicious Software)

- **Definition:** Software designed to damage, disrupt, or gain unauthorized access to systems.
- **Forms of Malware:**
 - **Virus:** Self-replicating; spreads by execution of an infected object (e.g., email attachment).
 - **Worm:** Self-replicating; spreads automatically by exploiting vulnerabilities, no user execution needed.
 - **Spyware:** Records user activity (keystrokes, visited sites, passwords) and sends it to attacker.
 - **Botnet:** A group of infected machines (bots) controlled remotely, used for spam or DDoS attacks.

(b) Denial of Service (DoS) Attacks

Definition:

A **DoS attack** is when someone **tries to block or slow down a website, server, or network** so that **normal users cannot access it**.

Mechanism:

1. The attacker sends a huge number of requests or data packets to the target.
2. The target’s resources (like bandwidth, memory, or CPU) become overloaded.

3. Normal users' requests cannot be processed, causing service disruption.
4. Sometimes, attackers take advantage of weaknesses in the system to make it crash or stop working.

(c) Packet Interception (Packet Sniffing)

Definition:

Packet interception, also called **packet sniffing**, is when someone **captures and reads data packets** traveling over a network without permission.

How it works (Mechanism):

- The attacker uses a **sniffer tool** to monitor network traffic.
- Data packets moving through the network are **captured**.
- Sensitive information like **passwords, emails, or personal data** can be **read or recorded**.

(d) Fake Identity (IP Spoofing)

Definition:

IP spoofing is when an attacker **pretends to be another computer or user** by **faking their IP address** to gain unauthorized access or trick a system.

How it works (Mechanism):

1. The attacker **changes the source IP address** in the packet header.
2. The target system **thinks the packet is from a trusted source**.
3. The attacker can then **bypass security checks, steal data, or launch attacks**.

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